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EXAMPLE 2.10

Compare the components for the following vector equations

a) $\vec{F} = m\vec{a}$ Here m is positive number

b) $\vec{p} = 0$

Solution

Case (a):

$$\vec{F} = m\vec{a}$$

$$F_x\hat{i} + F_y\hat{j} + F_z\hat{k} = ma_x\hat{i} + ma_y\hat{j} + ma_z\hat{k}$$

By comparing the components, we get

$$F_x = ma_x, F_y = ma_y, F_z = ma_z$$

This implies that one vector equation is equivalent to three scalar equations.

Case (b)

$$\vec{p} = 0$$

$$p_x\hat{i} + p_y\hat{j} + p_z\hat{k} = 0\hat{i} + 0\hat{j} + 0\hat{k}$$

By comparing the components, we get

$$p_x = 0,\ p_y = 0,\ p_z = 0$$

EXAMPLE 2.11

Determine the value of the T from the given vector equation.

$$5\hat{j} - T\hat{j} = 6\hat{j} + 3T\hat{j}$$

Solution

By comparing the components both sides, we can write

$$5 - 6 = 3T + T$$

$$-1 = 4T$$

$$T = -\frac{1}{4}$$

EXAMPLE 2.12

Compare the components of vector equation $\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = \vec{F}_4$

Solution

We can resolve all the vectors in x, y and z components with respect to Cartesian coordinate system.

Once we resolve the components we can separately equate the x components on both sides, y components on both sides, and z components on both the sides of the equation, we then get

$$F_{1x} + F_{2x} + F_{3x} = F_{4x}$$

$$F_{1y} + F_{2y} + F_{3y} = F_{4y}$$

$$F_{1z} + F_{2z} + F_{3z} = F_{4z}$$

2.6 POSITION VECTOR

It is a vector which denotes the position of a particle at any instant of time, with respect to some reference frame or coordinate system.

The position vector $\vec{r}$ of the particle at a point P is given by